المعهد الهنـدي للتكنولوجيـــا دلهي
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ABU DHABI

Cell Cycle and cell Division

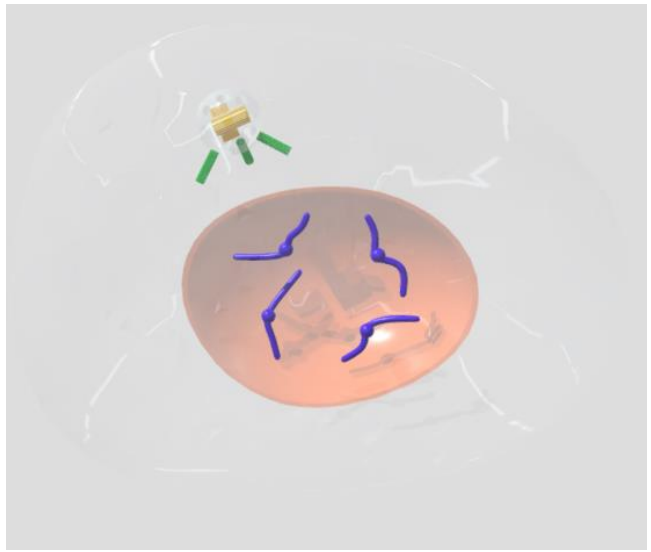
From unicellular to Multicellular

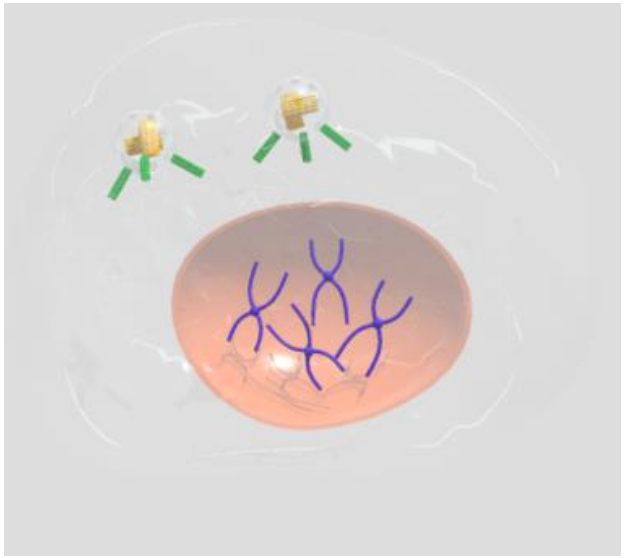
Lecture 21

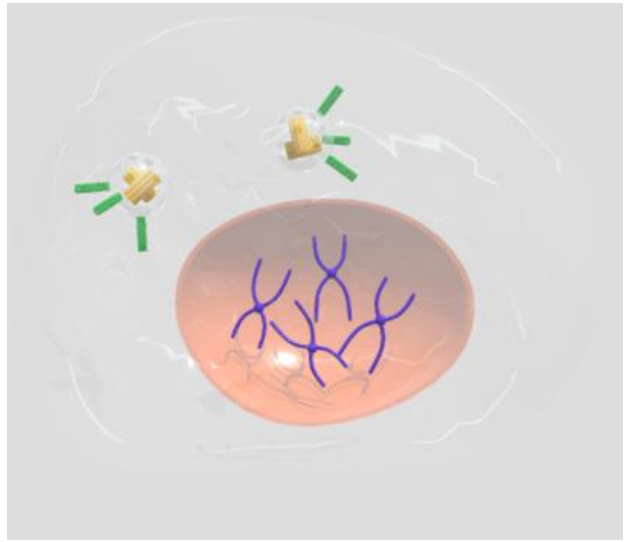
Prof. Saurabh Raj

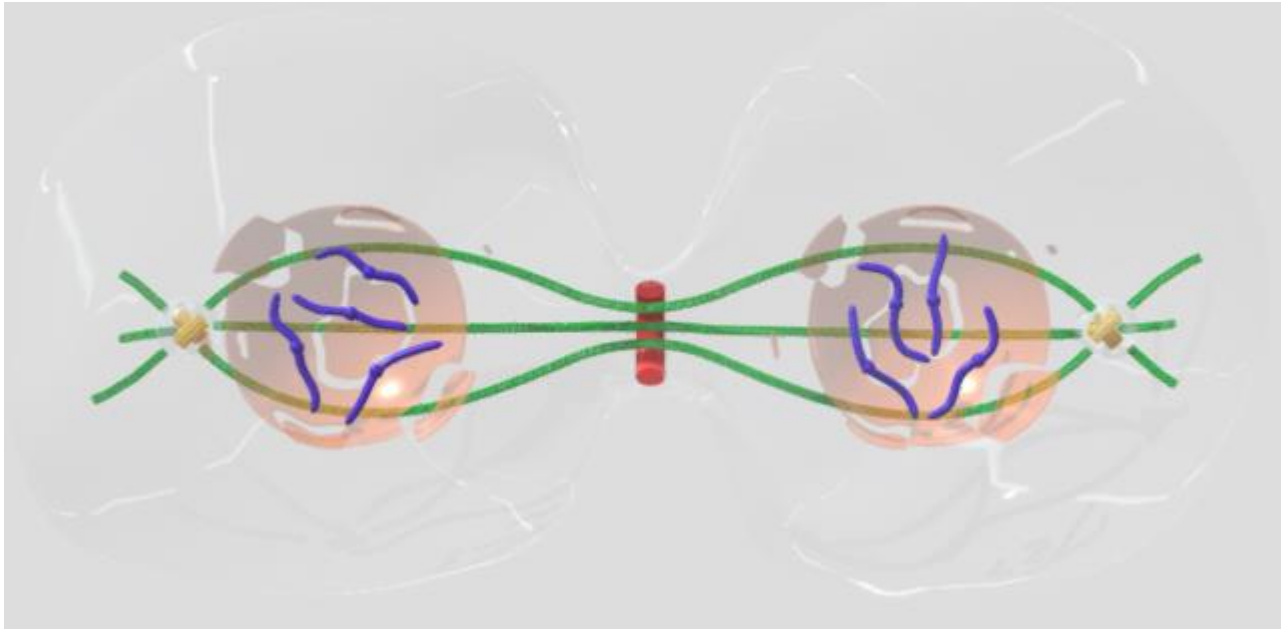
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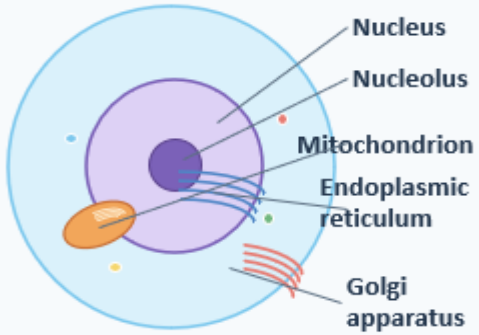




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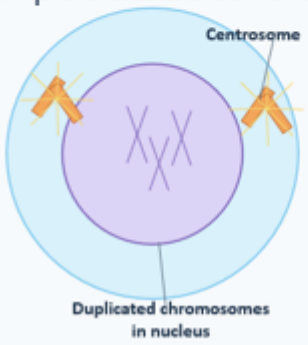
G1 Phase – Cell Growth

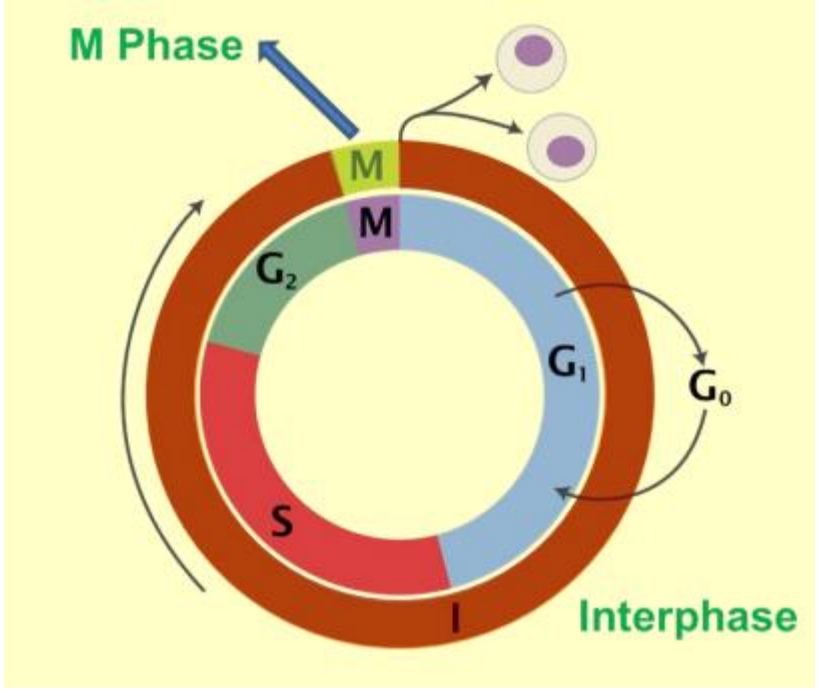
- Cell increases in size
- High metabolic activity
- Synthesis of RNA, proteins and organelles
- Monitors environmental conditions and internal signals
- Restriction point (G1 checkpoint) ensures the cell is ready to replicate DNA



G2 Phase

- Cell continues to grow
- Synthesis of proteins needed for mitosis (e.g., tubulin)
- Organelles are duplicated
- DNA is checked for errors and repairs are made
- G2 checkpoint ensures DNA replication is complete and the cell is ready for mitosis



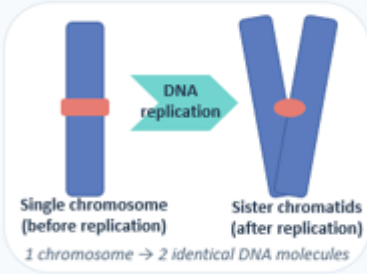


Interphase = G_1 + S + G_2

G_0 - Resting Phase

S Phase

- Entire genome is replicated once and only once
- Each chromosome is duplicated to form sister chromatids
- Histone proteins are synthesized
- DNA replication is highly accurate and proofread
- Checkpoint ensures replication is complete and DNA is undamaged

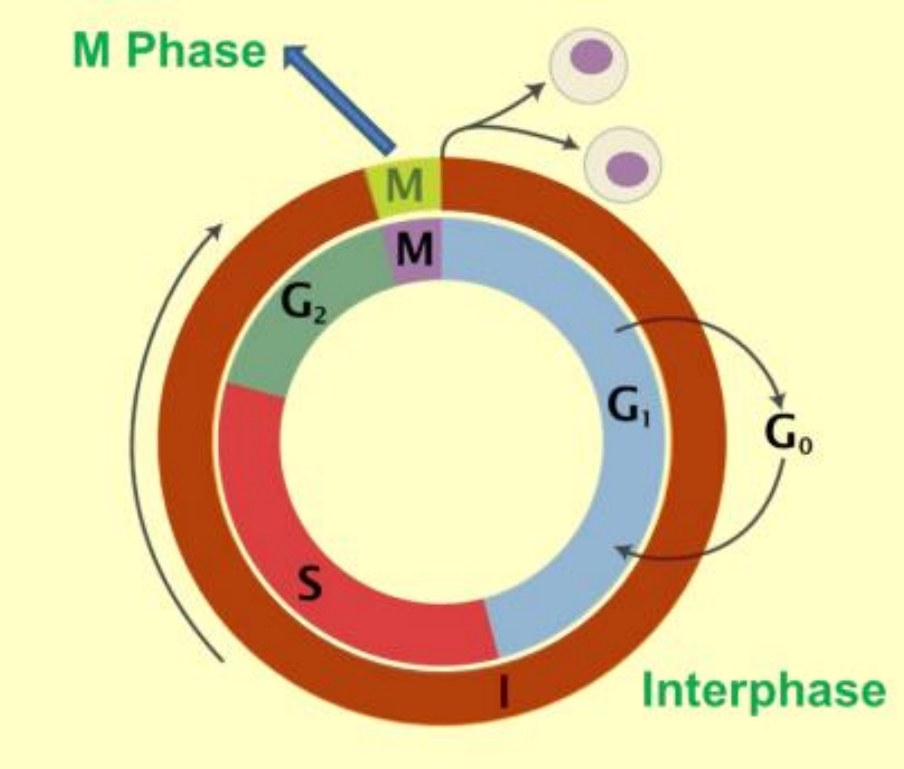


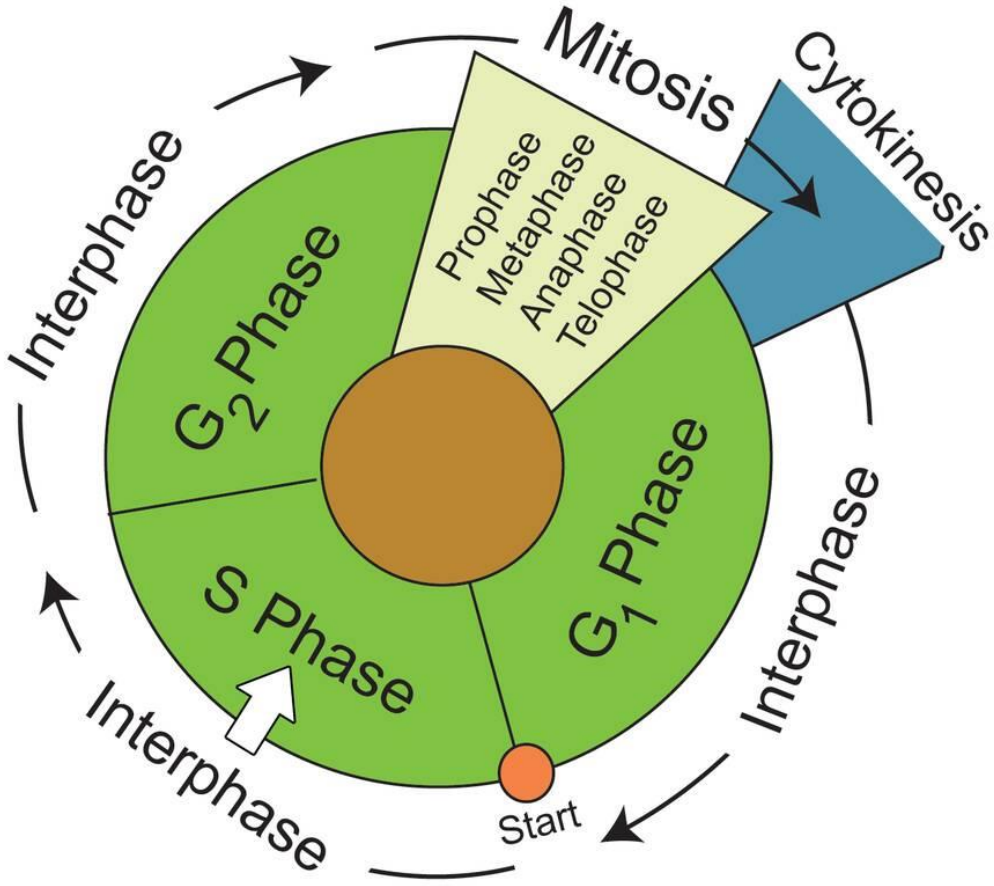
M Phase

- Mitosis: division of the nucleus into two genetically identical nuclei
- Cytokinesis: division of the cytoplasm to form two daughter cells
- Ensures equal segregation of chromosomes
- Produces two genetically identical diploid daughter cells



Mitosis


M Phase
M
M
G2
G1
G0
S
I
Interphase


Mitosis
Cytokinesis
Prophase
Metaphase
Anaphase
Telophase
Interphase
G2 Phase
S Phase
G1 Phase
Interphase
Interphase
Start

Mitosis: one division, two identical cells

Prophase

Chromosomes condense, The spindle forms and the nuclear nuclear envelope breaks down.

Metaphase

Chromosomes line up at the cell equator, attached to spindle fibres.

Anaphase

Sister chromatids separate and move to opposite poles. poles.

Telophase

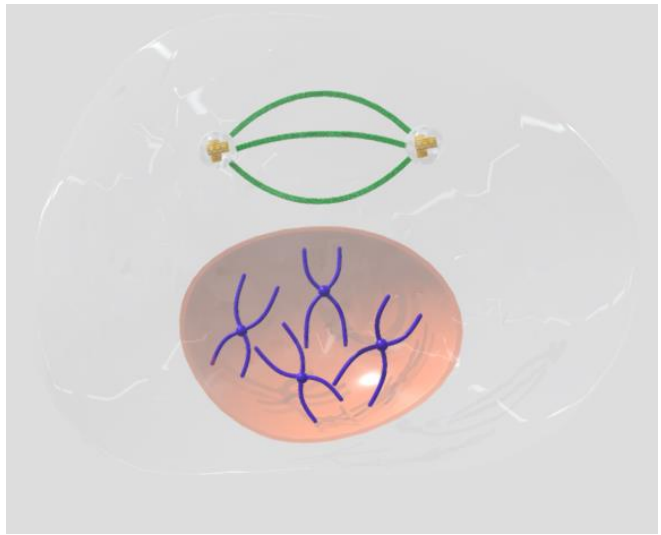
Nuclear envelopes re-form and chromosomes decondense.

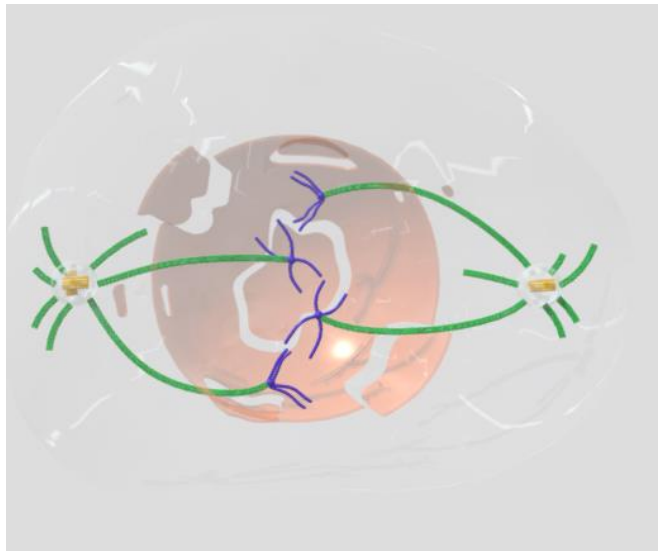
Cytokinesis

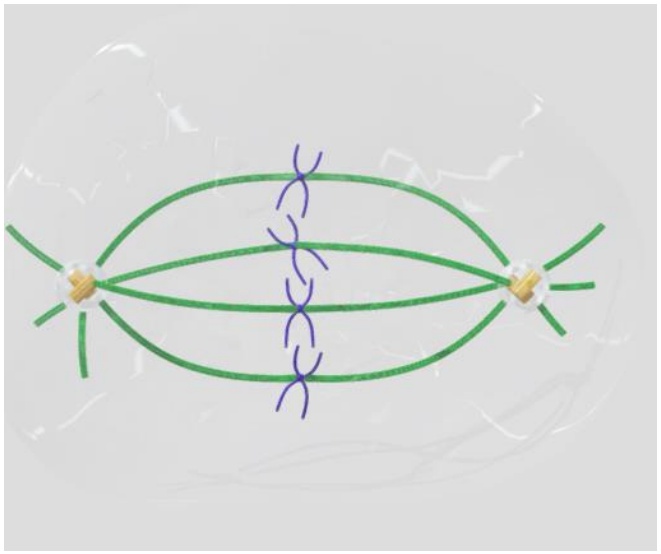
The cytoplasm divides: a cleavage furrow in animal cells, a cell plate in plant cells.

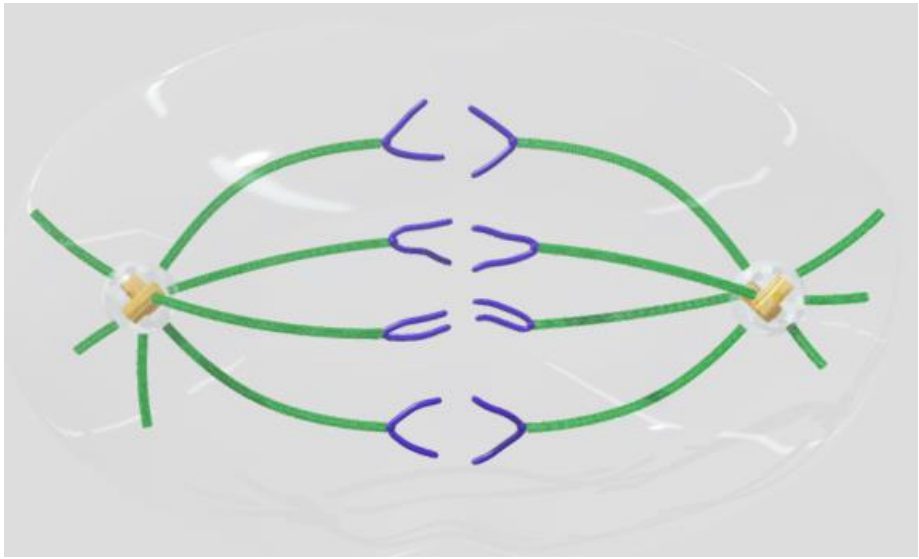
Result: two genetically identical diploid (2n) cells, used for growth, repair and asexual reproduction.

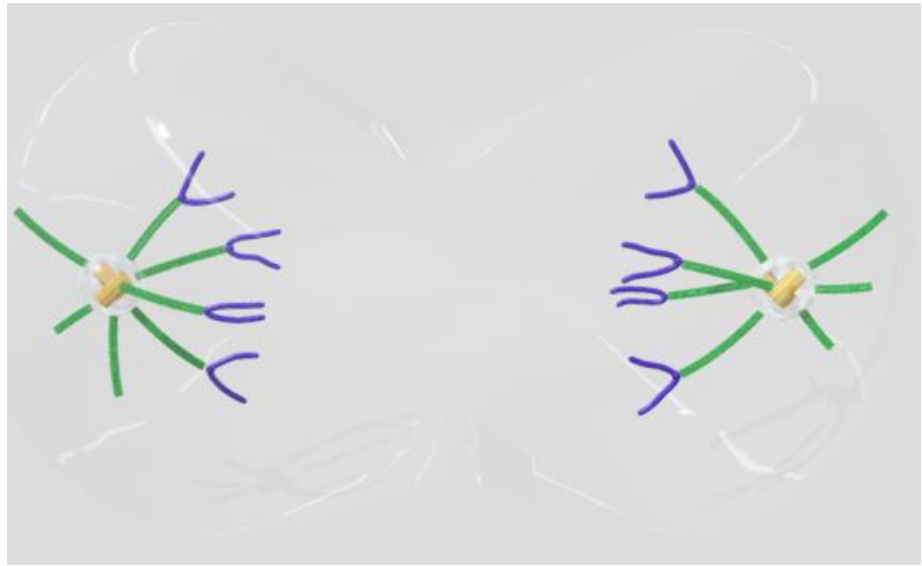
Mitosis in pictures





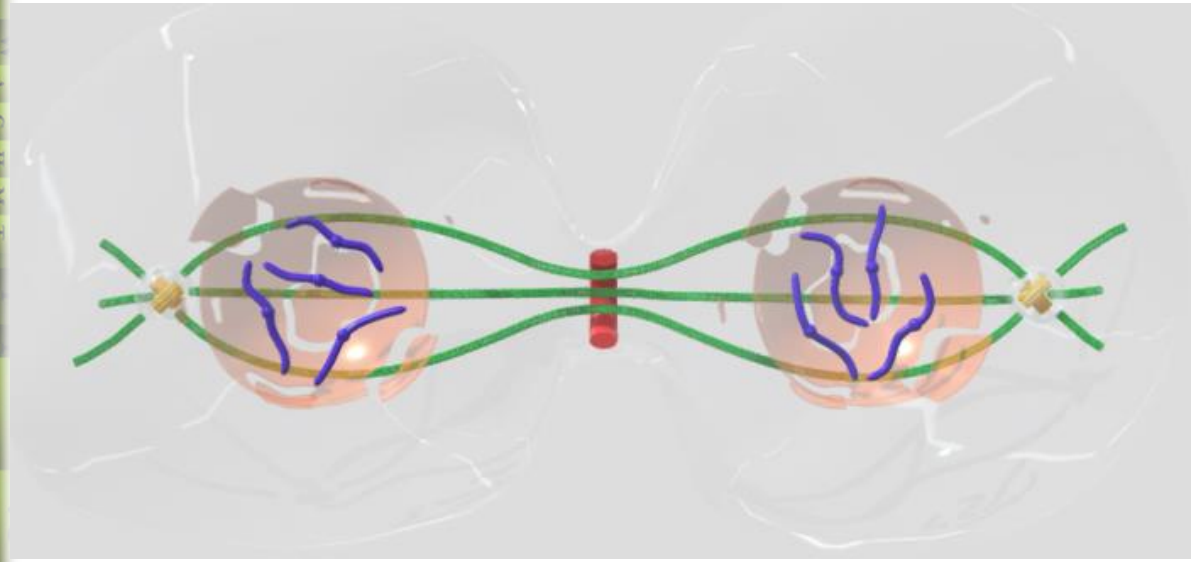


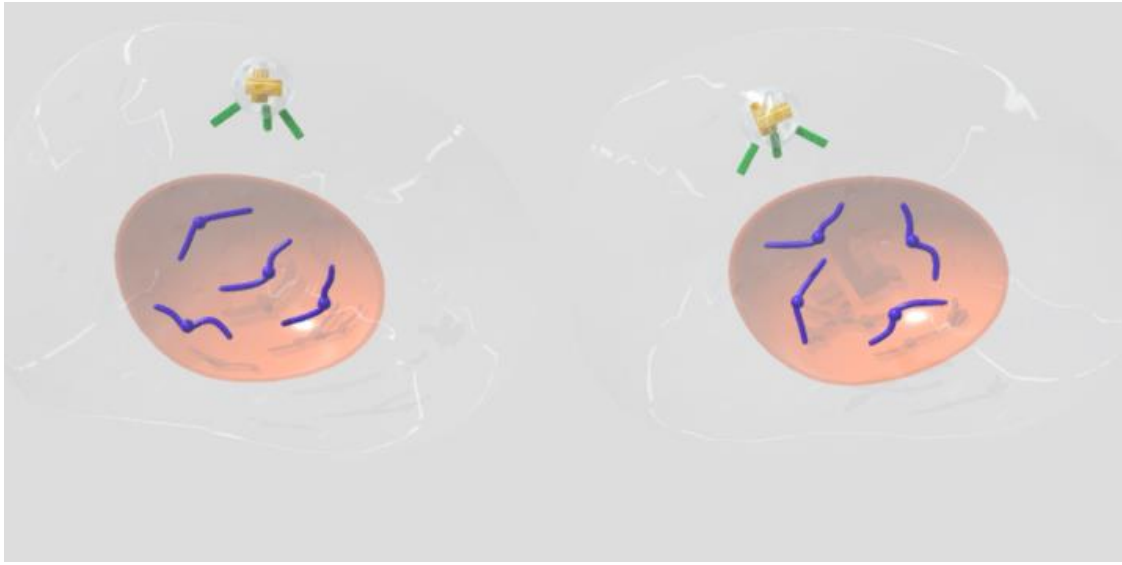




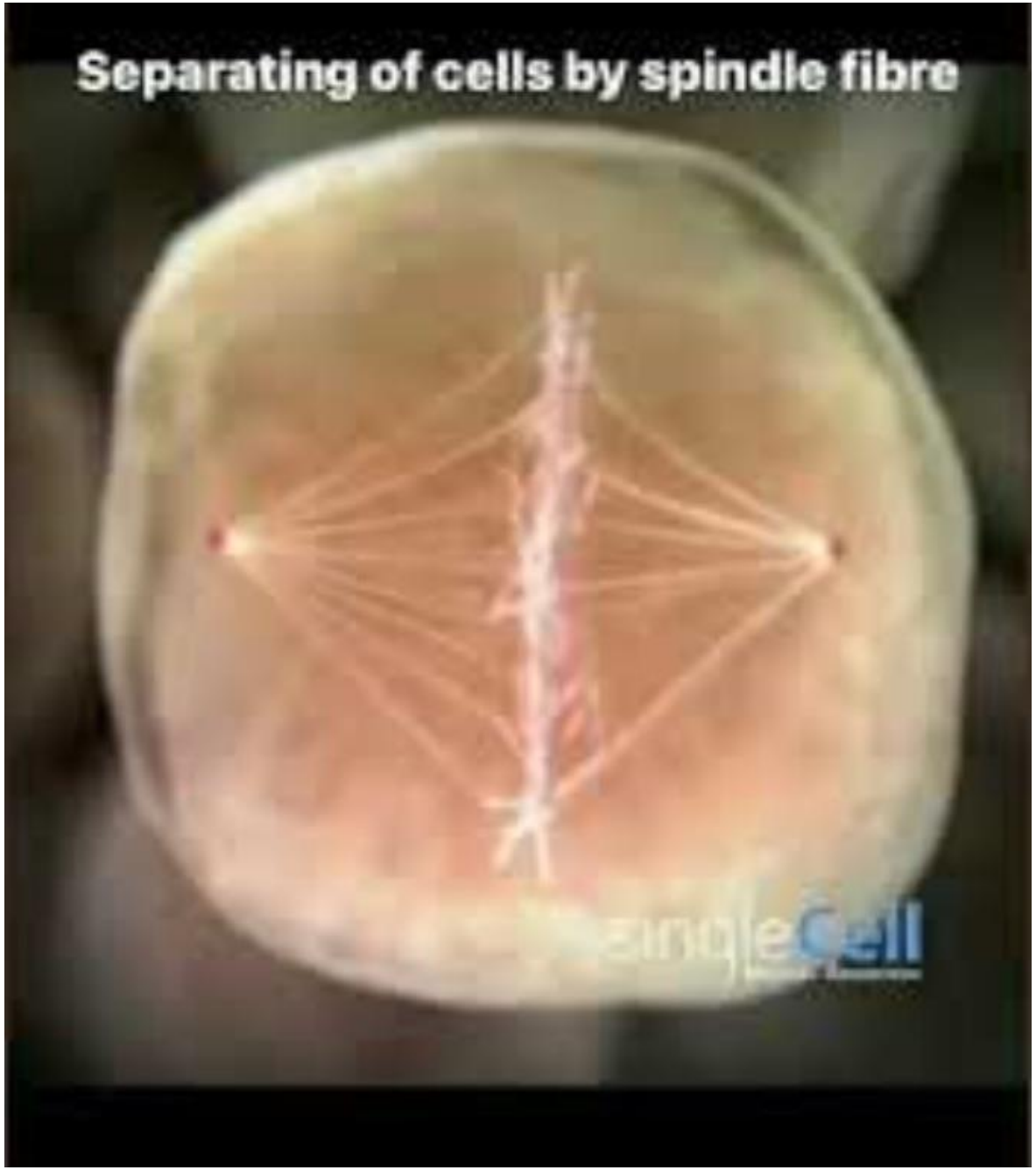
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Mitosis: one division, two identical cells



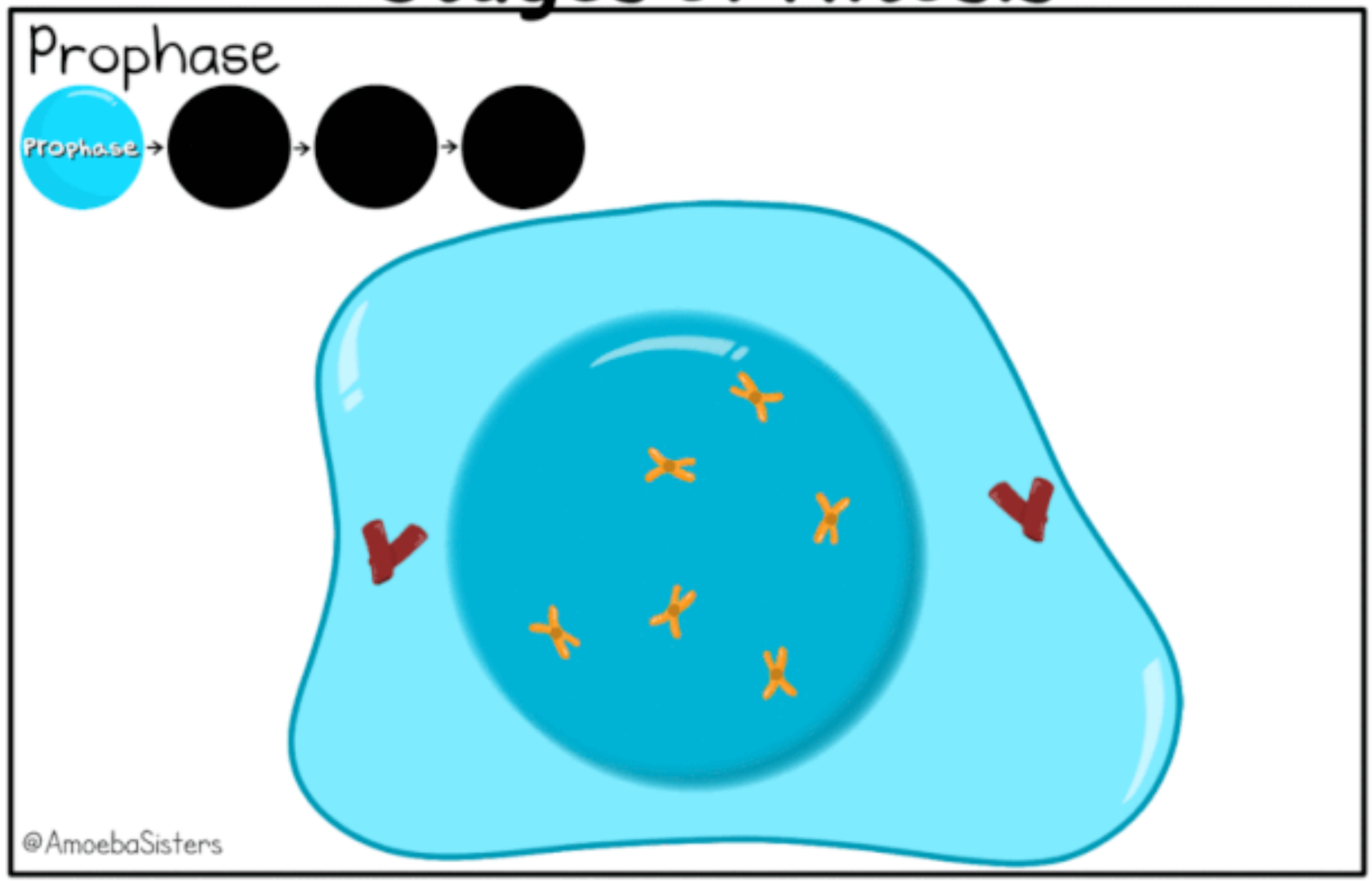


https://sketchfab.com/juliopanz/models



https://www.youtube.com/shorts/kn9iiwqFHTM

Stages of Mitosis



Checkpoints: pause, inspect, proceed

G_1 checkpoint

Is the cell large enough?

Are nutrients and growth signals available?

Is DNA damaged?

G_2 checkpoint

Has DNA replication finished?

Is the DNA sufficiently intact to enter mitosis?

Is the DNA replication error or damage free?

Spindle checkpoint

Are chromosomes correctly aligned and attached to spindle microtubules before separation?

Progression can be delayed when a problem is detected.

What is a kinase?

An enzyme that can change a protein's activity

ATP + target protein — kinase —▶ phosphorylated protein

1. Transfers phosphate

Usually transfers a phosphate group from ATP to a target molecule.

2. Changes behaviour

Phosphorylation may alter a protein's activity, shape, location or interactions.

3. Coordinates events

Kinases help regulate signalling, growth, DNA replication and cell division.

Key idea: phosphorylation can activate or inhibit a protein—it depends on the target.

CDKs: kinases that help time the cell cycle

CDK = cyclin-dependent kinase

CDK

A protein kinase.

It phosphorylates target proteins involved in cell-cycle events.

Cyclin

A regulatory protein.

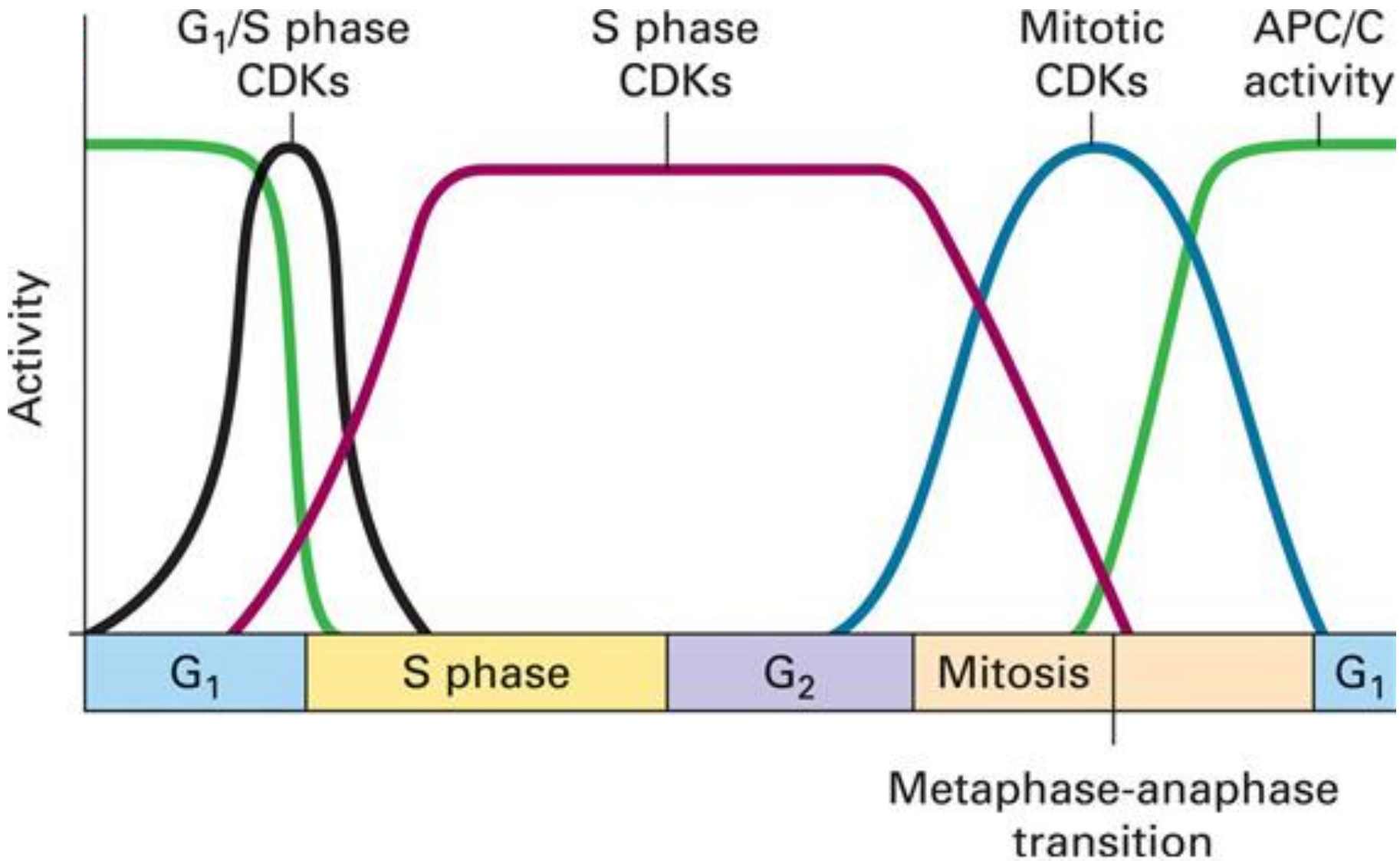
Cyclin levels rise and fall during the cell cycle and help regulate CDK activity.

Cyclin–CDK complex

The control unit.

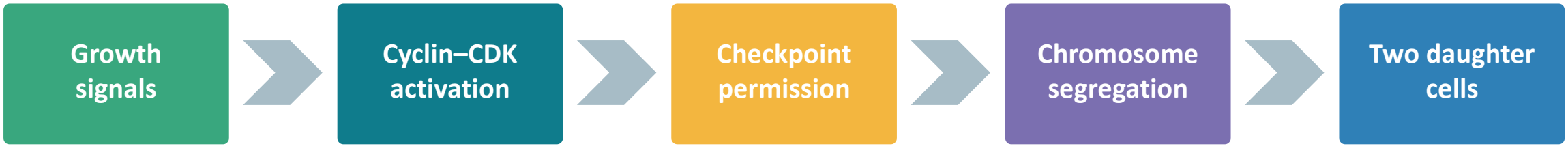
When appropriately activated, it phosphorylates selected targets to promote cell-cycle transitions.

Cyclin accumulates → binds CDK → CDK activity rises → target proteins are phosphorylated


G_1/S phase CDKs
S phase CDKs
Mitotic CDKs
APC/C activity
Activity
G_1
S phase
G_2
Mitosis
G_1
Metaphase-anaphase transition

Putting the logic together

Progression requires both a “go” signal and the absence of a “stop” signal.



If a checkpoint detects a problem, the sequence pauses until the problem is resolved or the cell adopts another fate.

Damage response: pause, repair, or exit

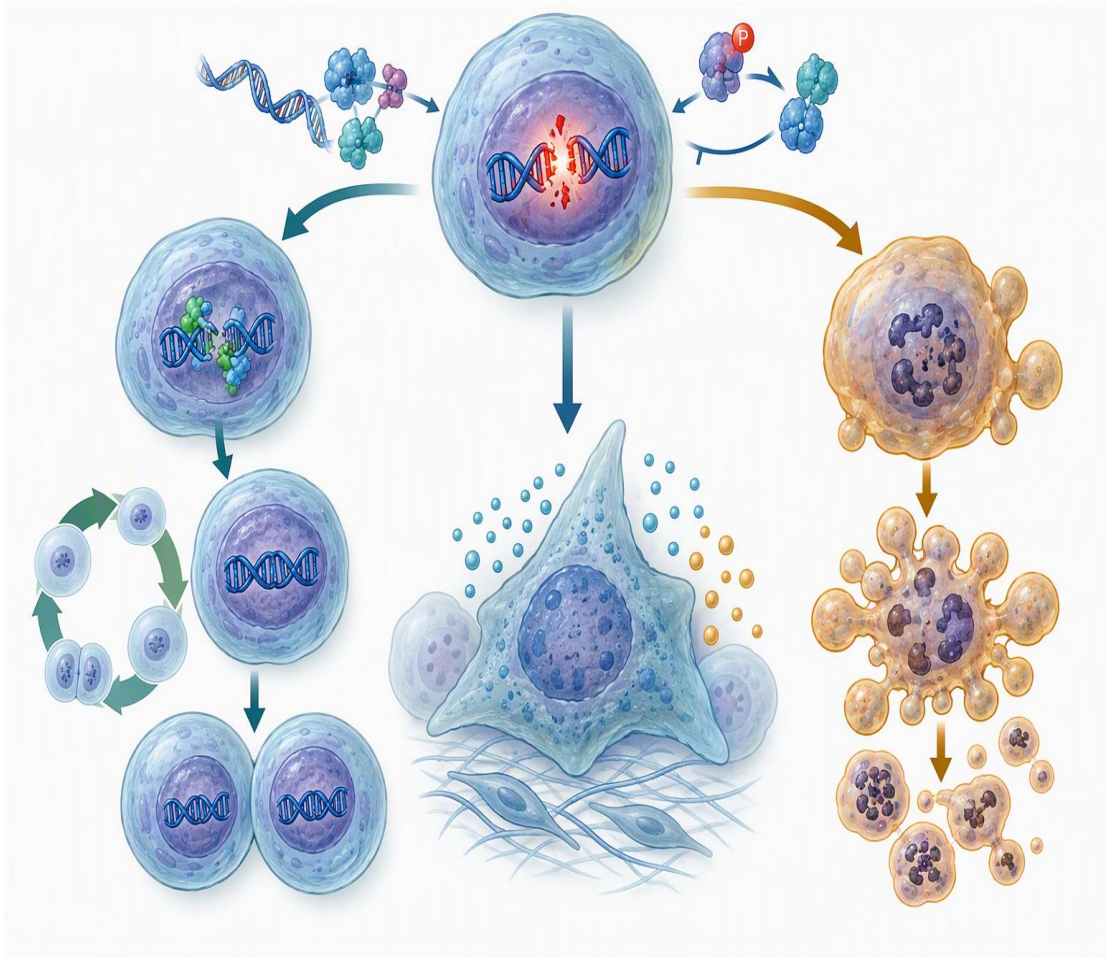
Checkpoint pathways convert molecular damage signals into cell-fate decisions.

DNA damage / replication stress

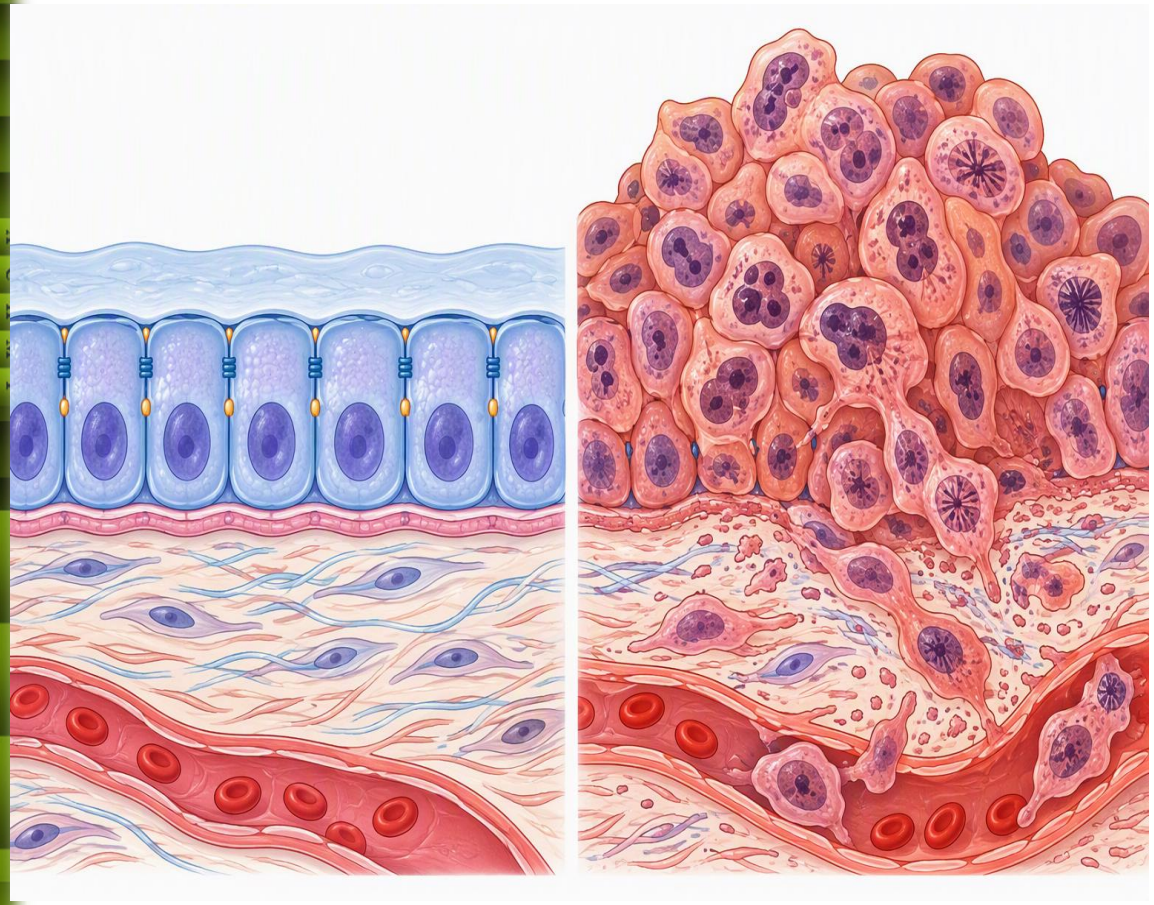
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Checkpoint signaling

Repair	Senescence	Apoptosis
resume cycle	stable arrest	eliminate cell

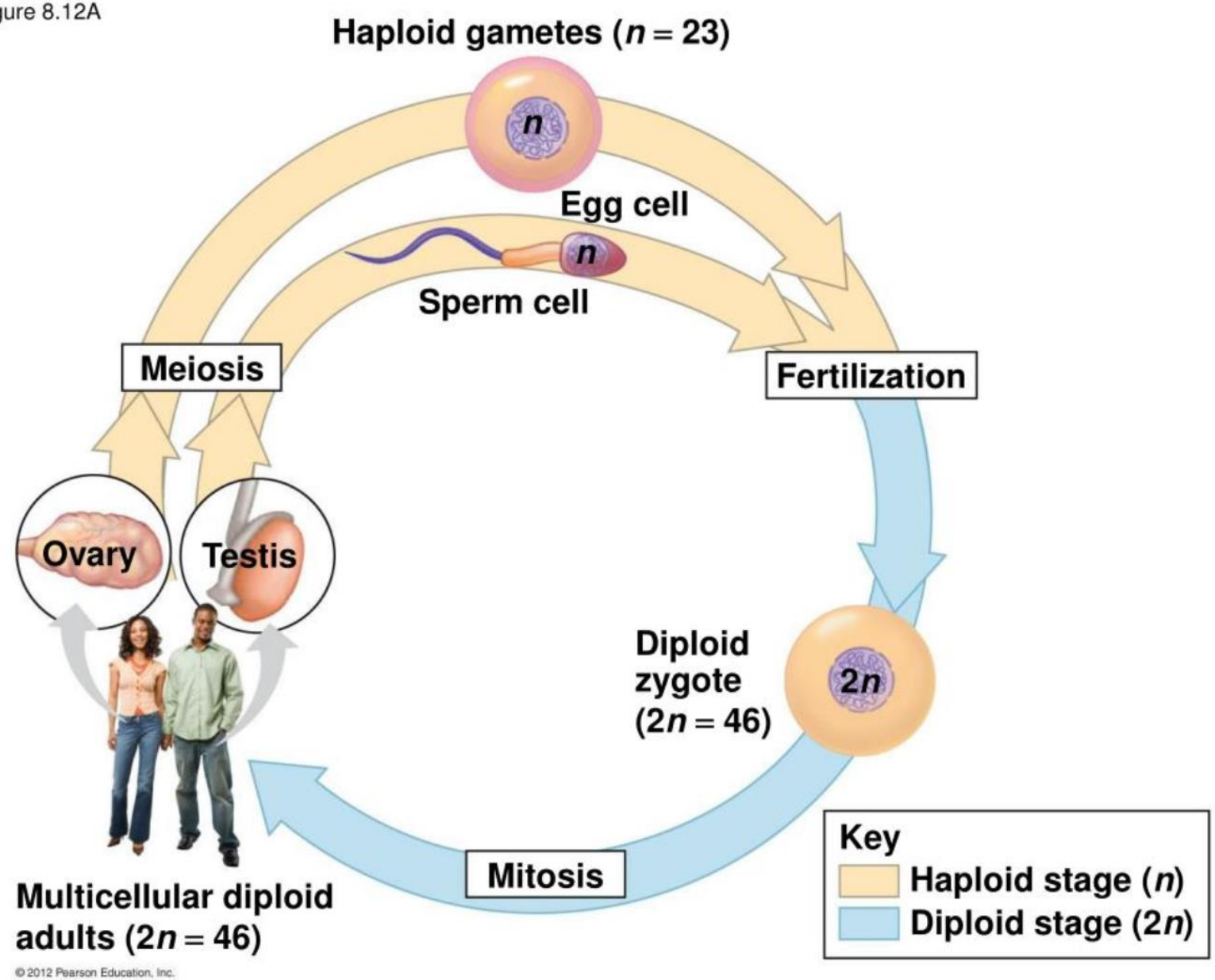


When control fails: the link to cancer



- Checkpoint defects allow damaged genomes to propagate
- Chromosome-segregation errors promote genomic instability
- Many anticancer strategies target cell-cycle dependencies

Figure 8.12A

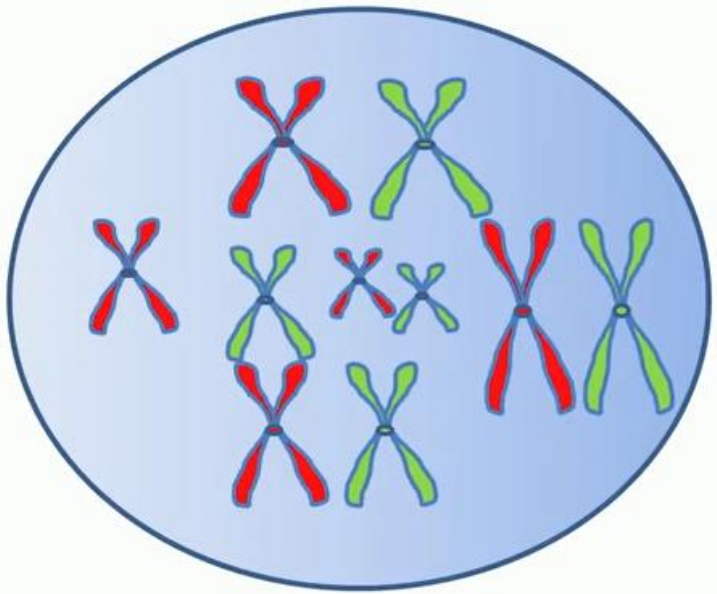


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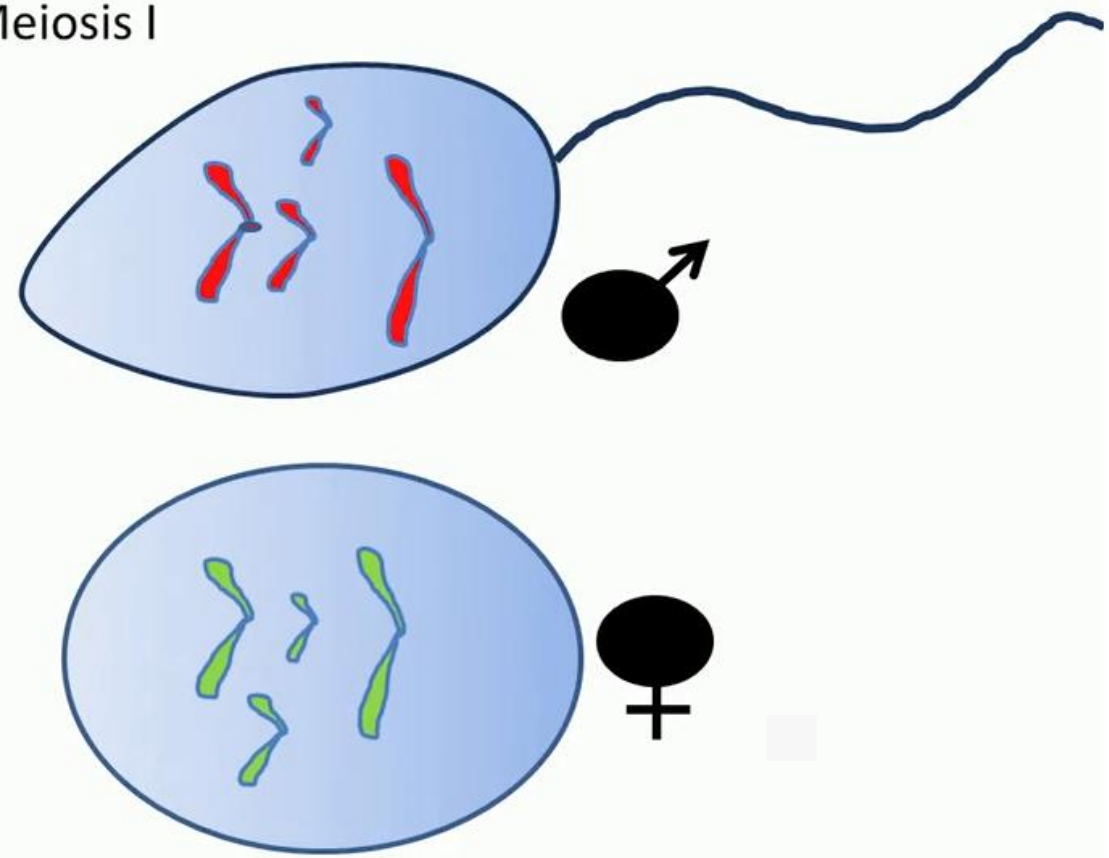
Diploid cells

Separation of Homologus chromosomes: Meiosis I

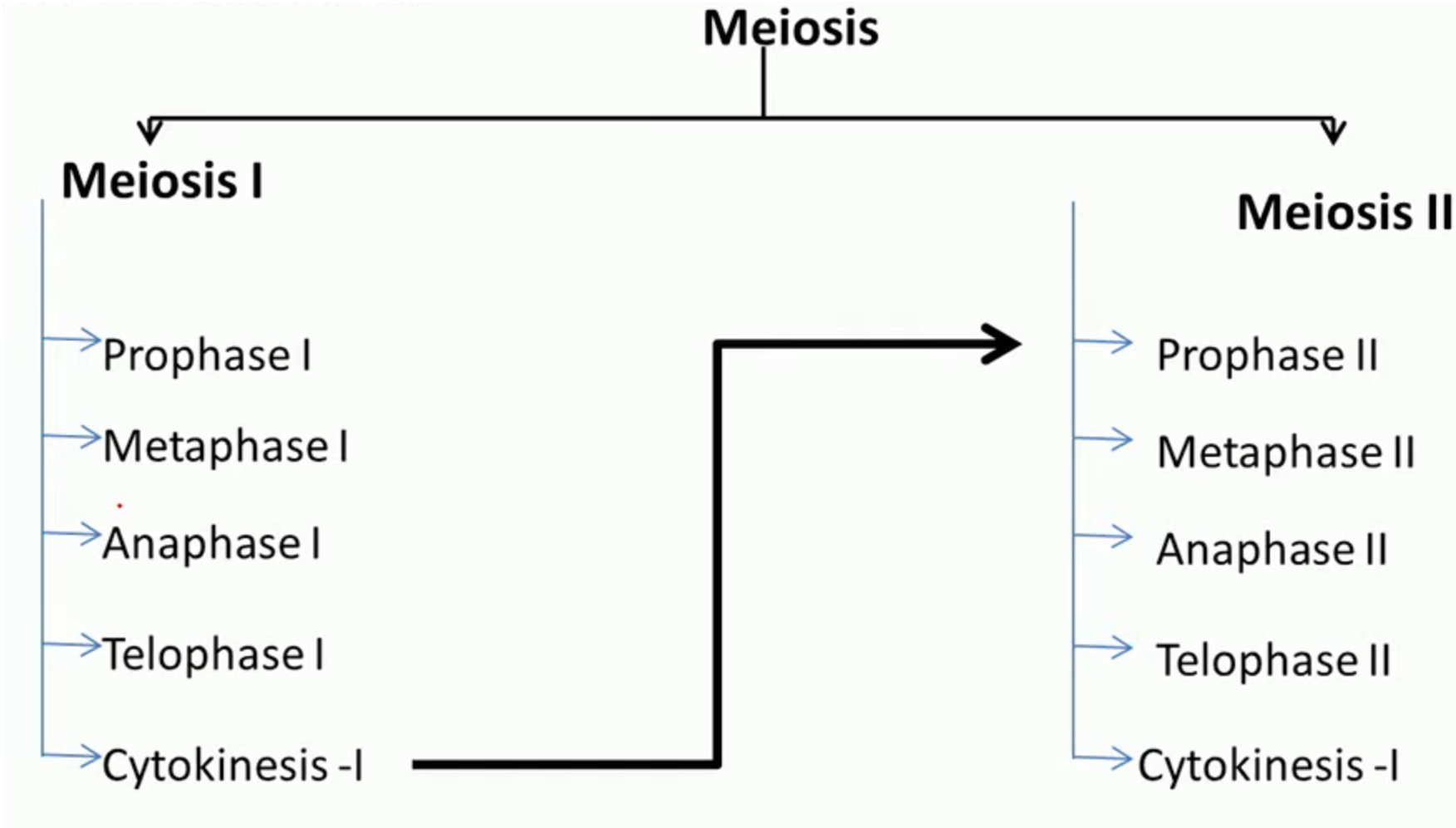
Separation of sister chromatids: Meiosis II



Diploid cell
(2n)



Haploid gametes
(1n)


Meiosis
Meiosis I
Meiosis II
Prophase I
Metaphase I
Anaphase I
Telophase I
Cytokinesis -I
Prophase II
Metaphase II
Anaphase II
Telophase II
Cytokinesis -I

Meiosis: two divisions, four cells

1 cell

Diploid (2n), DNA already replicated.

Meiosis I →

2 cells

Haploid (n), chromosomes still doubled.

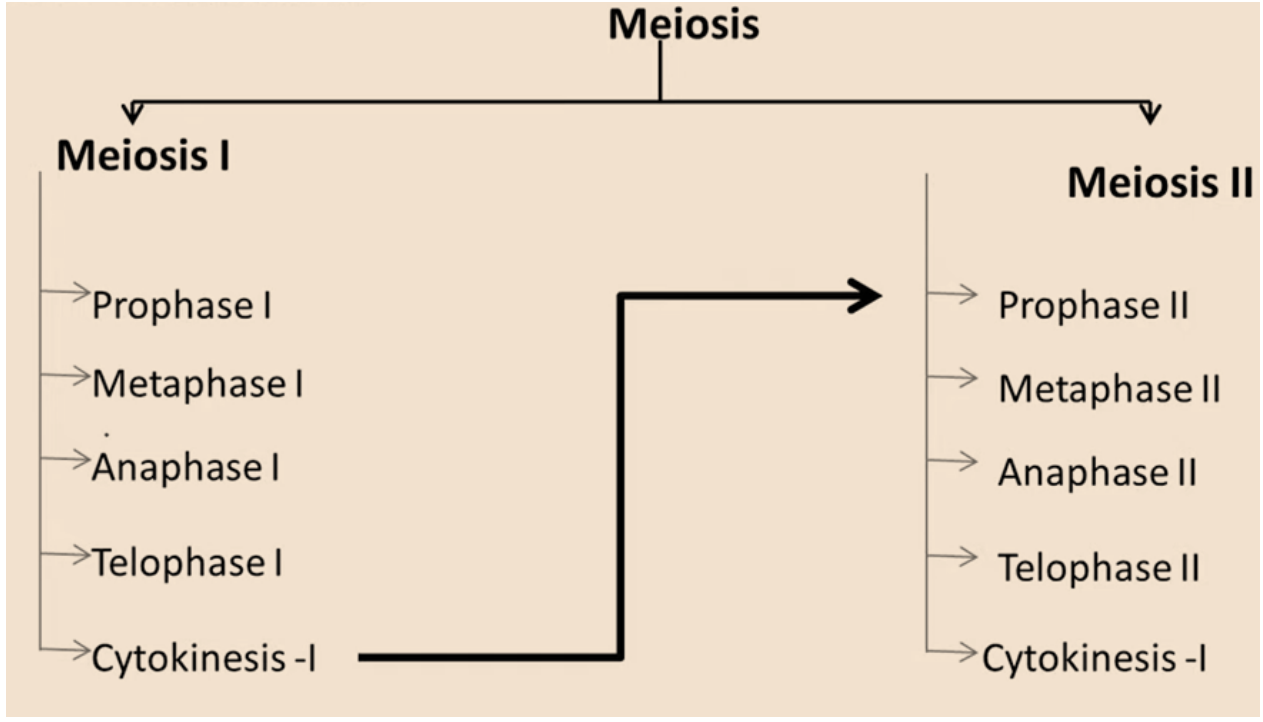
Meiosis II →

4 cells

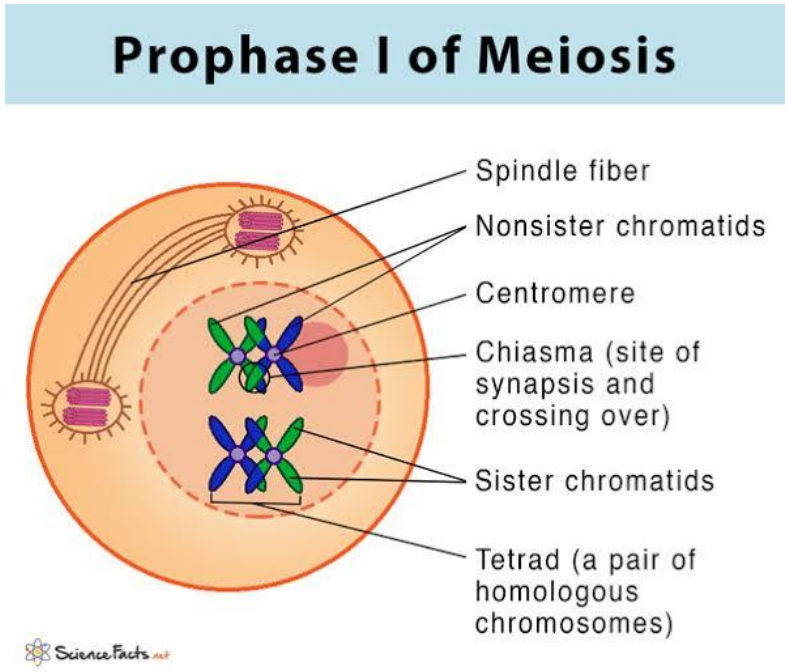
Haploid (n), each genetically unique.

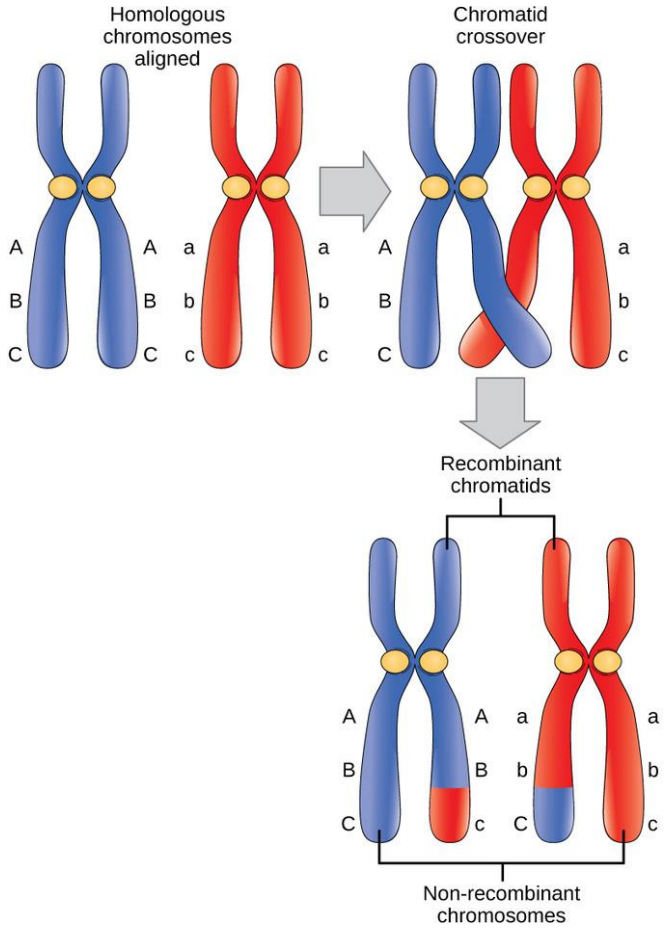
Meiosis I: homologous chromosomes pair, cross over and separate.

Meiosis II: sister chromatids separate, much as in mitosis.



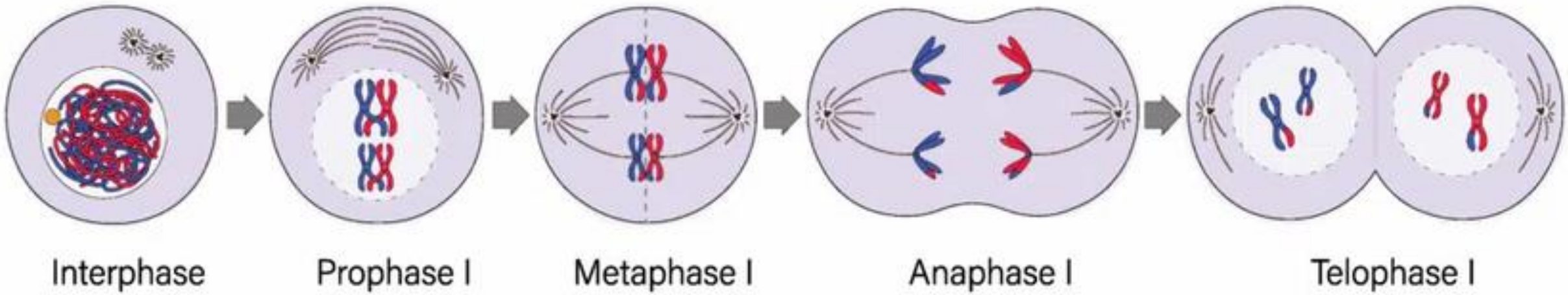
Prophase I



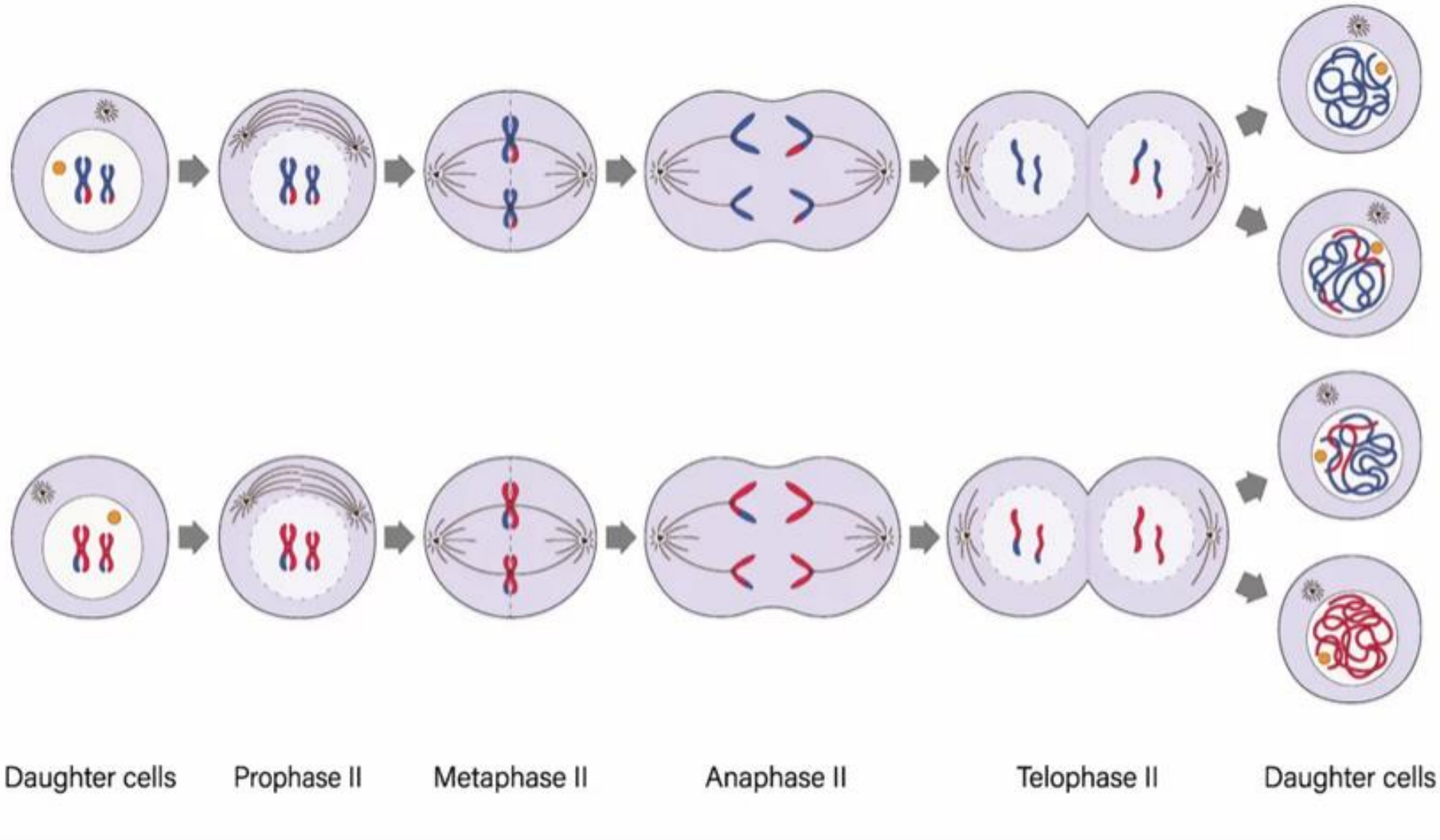


- Longest phase of Meiosis
- Spindle Formation
- Nuclear envelope starts to break
- Homologous chromosomes start to pair
- Cross over (Synapse)
- Genetic Variation

Meiosis I - Phases



Meiosis II - Phases



Variation, and when division goes wrong

Crossing over

In prophase I, homologous chromosomes swap segments, creating new allele combinations.

Independent assortment

Each homologous pair lines up independently at metaphase I, giving about 8.4 million chromosome combinations in humans.

Random fertilization

Any sperm can fuse with any egg, multiplying the possible outcomes further.

Nondisjunction

Chromosomes fail to separate, giving gametes with extra or missing chromosomes, e.g. trisomy 21 (Down syndrome).

Lost checkpoint control

Damaged cells keep dividing by mitosis, a hallmark of cancer.

Mitosis compared with meiosis

	Mitosis	Meiosis
Purpose	Growth, repair, asexual reproduction	Gametes for sexual reproduction
Occurs in	Somatic cells	Germ Cells
Divisions	One	Two
Daughter cells	Two	Four
Chromosome number	Diploid (2n), unchanged	Haploid (n), halved
Genetic identity	Identical to parent cell	Genetically unique